

Configuring Web Search Capabilities

Why Configure Web Search Capabilities?

While OpenClaw's local models are powerful, their knowledge is static, containing only the data available at the time of their training. This means:

- ✗ They cannot answer questions about the latest news and events.
- ✗ They do not know what the weather is like today.
- ✗ They cannot look up specific information on demand.
- ✗ They cannot access the most up-to-date technical documentation.

By configuring web search capabilities, OpenClaw can:

- ☒ Search the internet for information in real-time.
- ☒ Retrieve the latest news and updates.
- ☒ Query real-time data (weather, stock prices, etc.).
- ☒ Access the newest technical documentation and tutorials.
- ☒ Conduct in-depth research and fact-checking.

In this section, we will configure OpenClaw's web search capabilities. OpenClaw supports various search engines; the default `web_search` tool utilizes the Brave API. However, as of the publication of this tutorial, the Brave API has discontinued its free tier; therefore, we will use Tavily Web Search as an alternative.

Tavily Web Search is a search engine specifically optimized for AI agents, providing structured search results.

Prerequisites

Before you begin, please ensure the following:

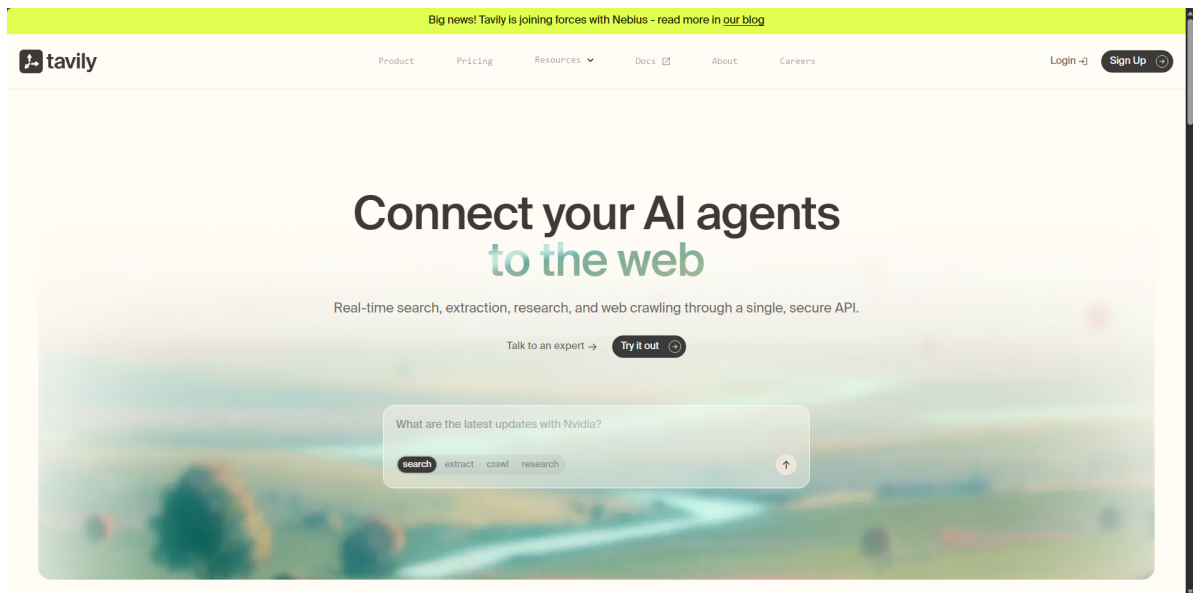
1. ☒ OpenClaw is installed and running correctly.
2. ☒ At least one model (either a local Ollama model or a cloud-based model) has been configured.
3. ☒ You have an active internet connection.
4. ☒ You have a valid email address (required to register for Tavily).

Step 1: Obtain a Tavily API Key

Tavily is an AI-optimized search engine that offers a free usage tier, providing 1,000 free searches per month—which is more than sufficient for typical daily use. (Policies are subject to change; specific quotas are based on actual limits.)

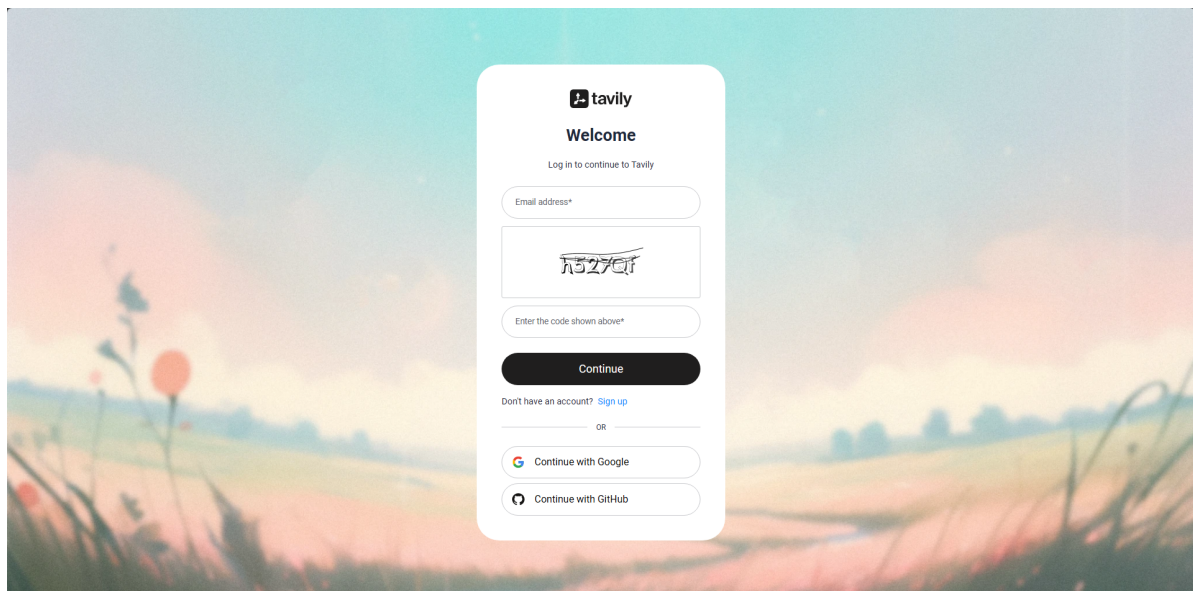
1.1 Visit the Tavily Website

Open the following URL in your browser: <https://tavily.com>



1.2 Sign Up for an Account

Click the "Sign Up" or "Register" button on the page:



Fill in your registration details:

- **Email Address:** Enter your valid email address.
- **Code:** The graphical verification code.

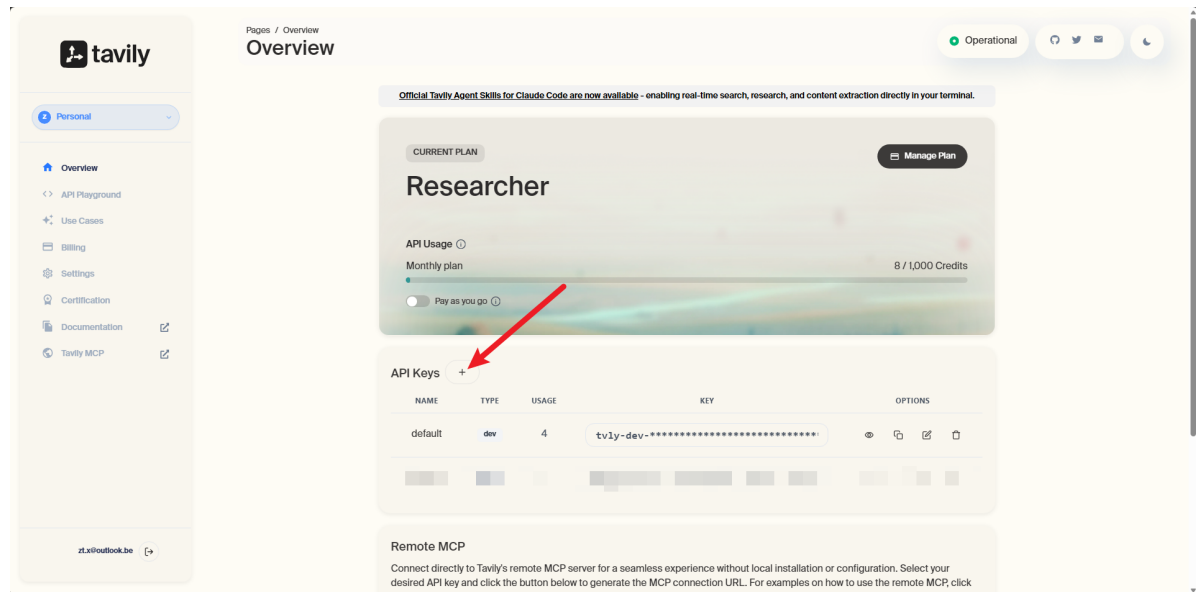
Click "Continue" to create your account.

1.3 Verify Your Email

Tavily will send a verification email to your inbox. Open your email client, locate the email from Tavily, and click the verification link. Once successfully verified, you will be automatically redirected to the Tavily dashboard.

1.4 Obtain Your API Key

After logging in, navigate to the "API Keys" page:



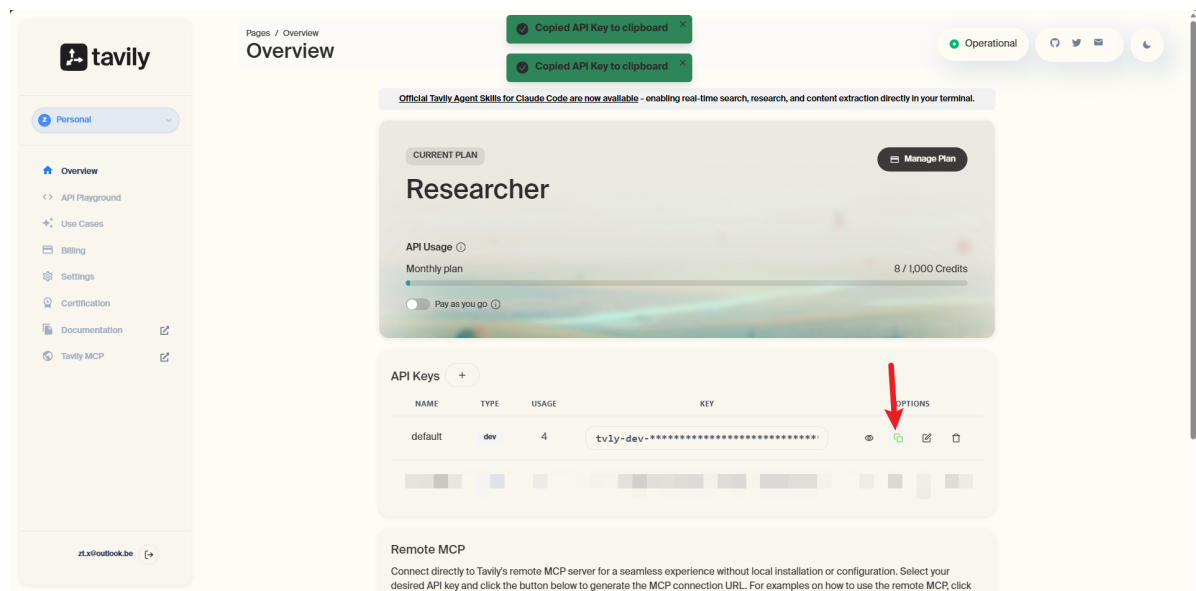
You will see an API key displayed, formatted similarly to this:

[illegible]

⚠ Important Note:

- Please keep this key secure and do not share it with anyone else.
- Do not commit this key to code repositories or public platforms.

Copy this API key; we will need it later.



Step 2: Install the Tavily Web Search Skill

OpenClaw uses ClawHub to manage skills; we can install this plugin directly by interacting with OpenClaw via chat.

2.1 Open the Terminal

Open a terminal window on your Jetson device. (For the Jetson Nano, you must enter the Docker container environment.) Ensure that you have a terminal window open with the `openclaw gateway` running.

2.2 Installing Skills

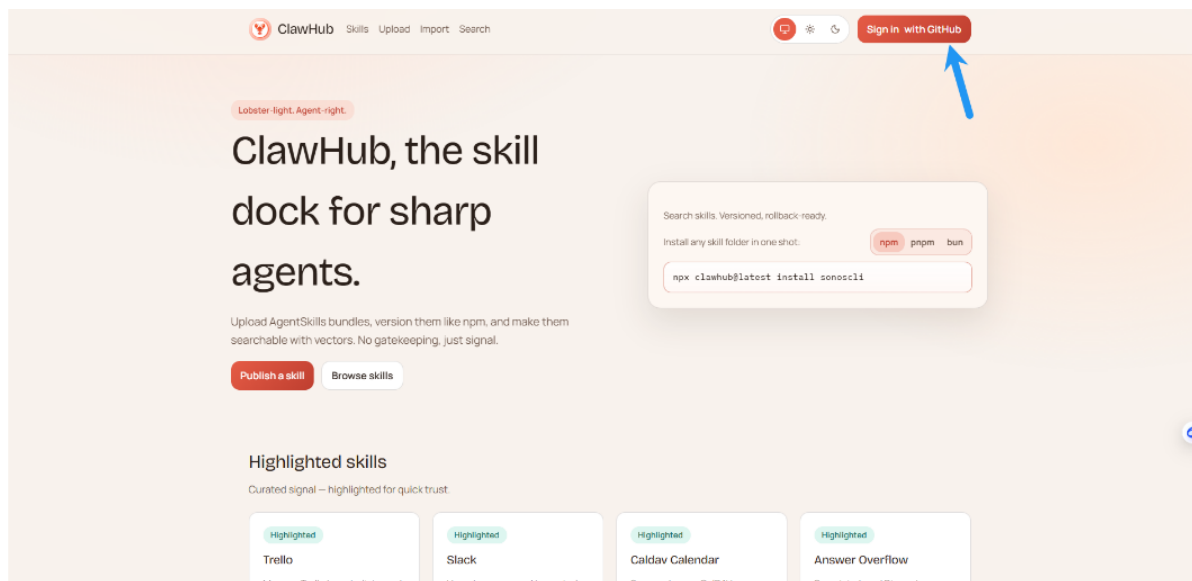
Before installing any skills, you first need to log in to ClawHub. This is an open-source distribution hub for OpenClaw skills—think of it as an "App Store" for your mobile device.

If you do not log in, you may encounter download speed limitations when attempting to download skills.

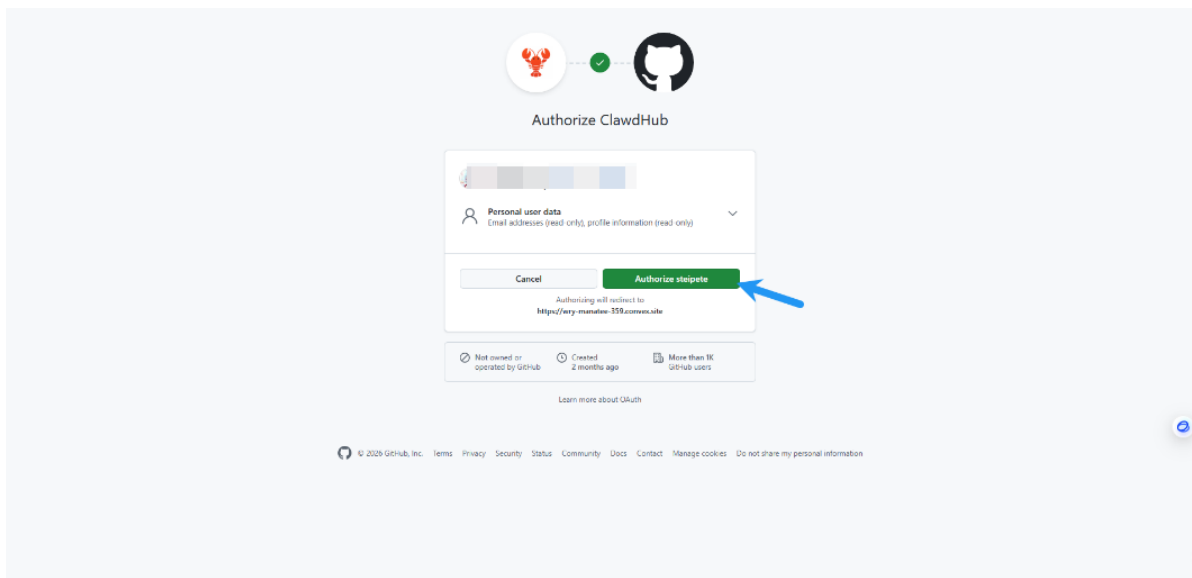
Open a terminal window and enter the following command:

```
npm i -g clawhub
```

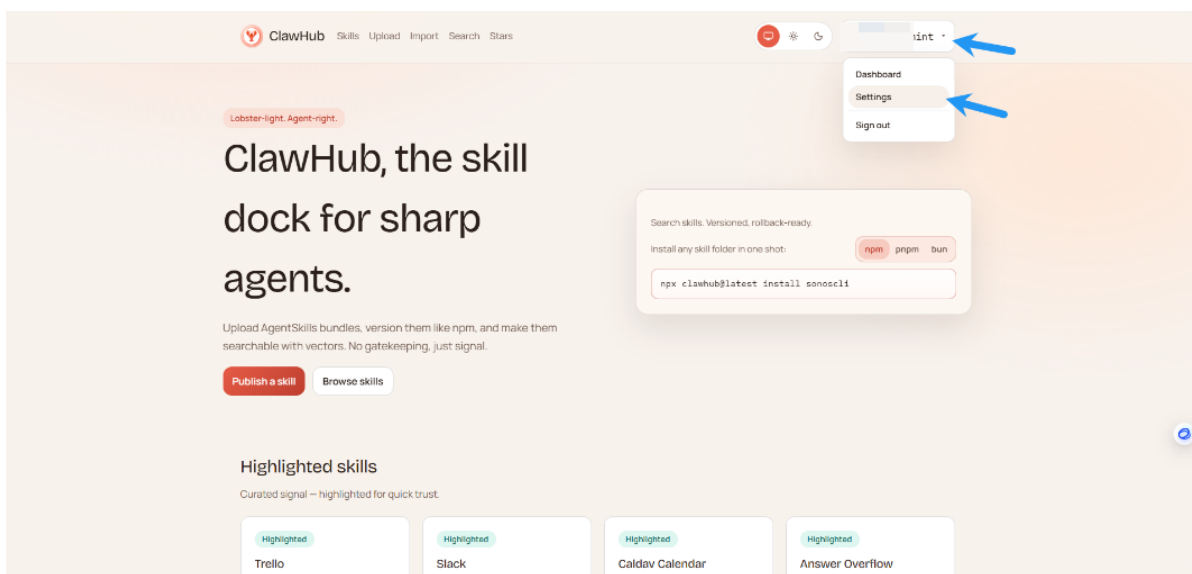
Next, visit the official ClawHub website at <https://clawhub.ai/> and click to log in using your GitHub account. (You must be logged into GitHub beforehand; we will not demonstrate that step here. To ensure you can proceed even if you encounter network issues preventing access to GitHub, we have pre-installed this specific skill in the factory-provided system image. You may therefore skip directly to Step 3: Configuring API Keys.)



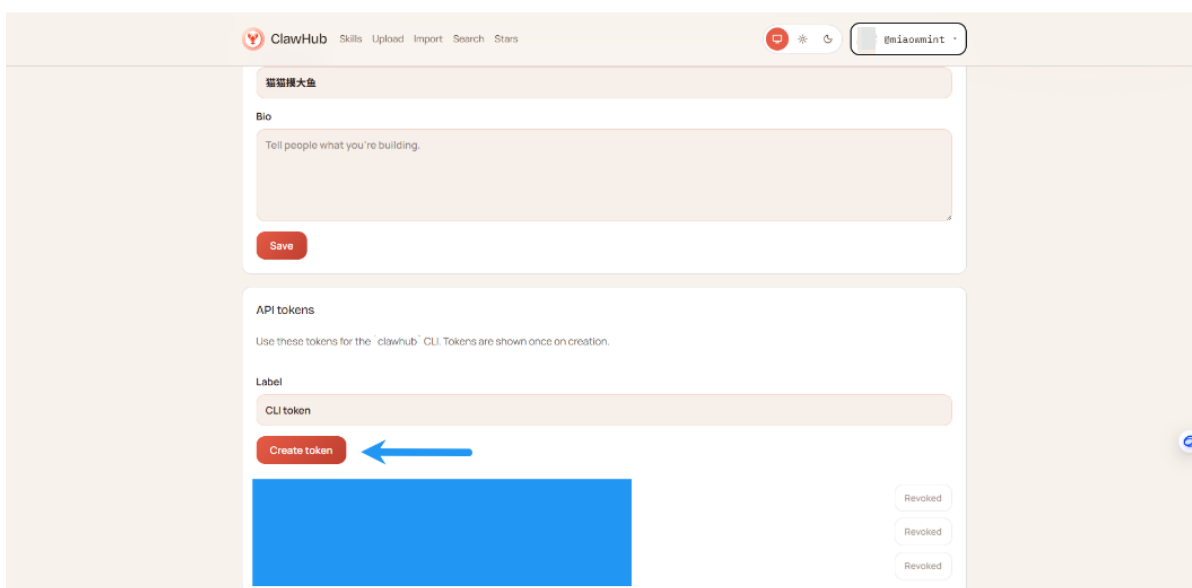
You will be redirected to an authorization page; click "Authorize" to proceed.



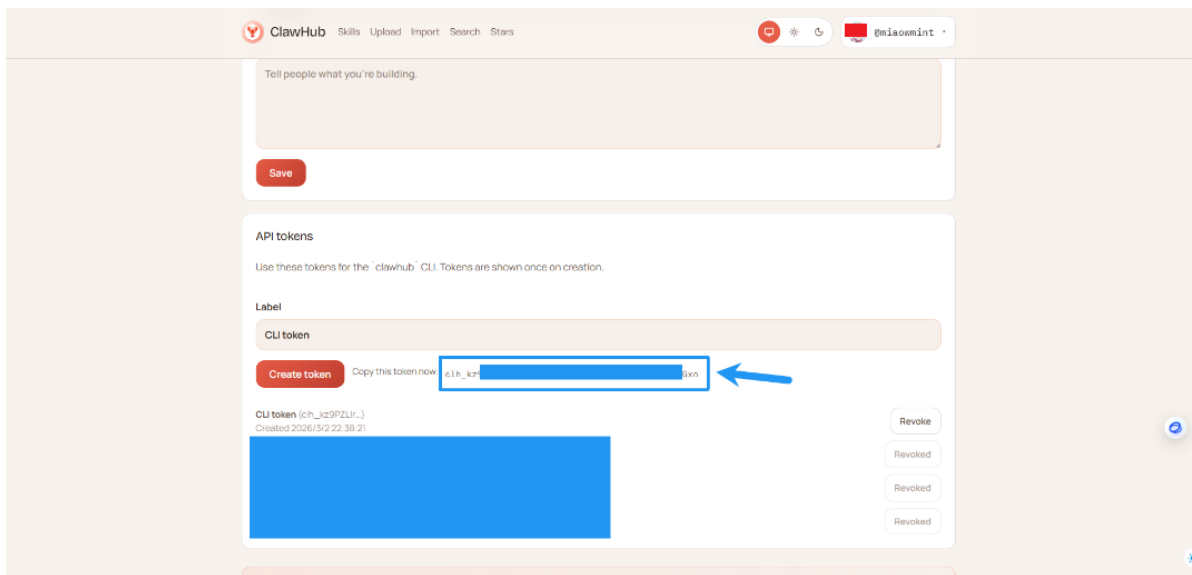
Click on your username, then click "Settings."



Scroll down to the "API tokens" section and click "Create token."



Copy the generated token. —which you will need to use below:



Next, run the following command on your server, replacing the placeholder with the Token you just copied. Note that there is a space following `--token`.

```
clawhub login --token YourToken
```

You should see a confirmation indicating that the login was successful:

```
root@jetson-yahboom:/# npm i -g clawhub
changed 35 packages in 3s
20 packages are looking for funding
  run `npm fund` for details
root@jetson-yahboom:/# clawhub login --token clh_...otIJc
✓ OK. Logged in as @zt
root@jetson-yahboom:/#
```

Next, we need to install the `tavily-web-search` Skill. While there are several ways to do this, we will use a common, general-purpose method here: enter the following command in your terminal:

```
npx clawhub@latest install tavily-search
```

If you encounter a `✗ Rate limit exceeded` error, it indicates that there was an issue with the login step you performed earlier; please attempt to log in again.

2.3 Verifying the Installation

To view a list of your installed skills, run the following command:

```
npx clawhub@latest list
```

```
root@jetson-yahboom:~# clawhub list
taviy-search 1.0.0
root@jetson-yahboom:~#
```

You should now be able to see `taviy-web-search` listed among the installed skills.

Step 3: Configure API Key

Now we need to configure the Tavily API key within OpenClaw.

3.1 Method 1: Temporary Configuration via Environment Variables

Set the environment variable in your terminal:

```
export TAVILY_API_KEY=tvly-your-api-key-here
```

Note: Replace `tvly-your-api-key-here` with the actual API key you obtained in Step 1.

3.2 Method 2: Making Environment Variables Permanent

If you want the API key to be set automatically every time you launch your terminal, you can add it to your shell configuration file.

For Bash Users:

```
echo 'export TAVILY_API_KEY=tvly-your-api-key-here' >> ~/.bashrc
source ~/.bashrc
```

3.3 Apply Changes

Restart the OpenClaw Gateway to apply the configuration:

```
openclaw gateway restart
```

Step 4: Test the Web Search Functionality

Now, let's test whether the web search feature is working correctly.

4.1 Test Using the TUI

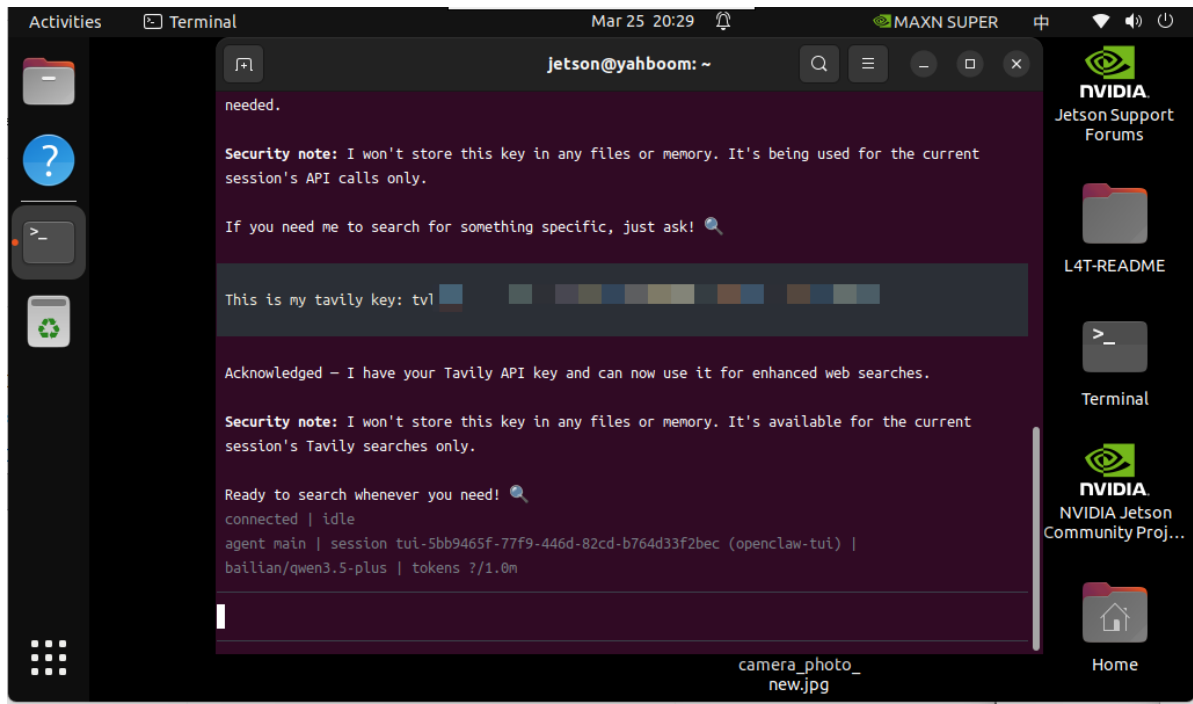
Launch the TUI in your terminal:

```
openclaw tui
```

```
root@jetson-yahboom:~# openclaw tui
w OpenClaw 2026.3.8 (3caab92) - I'm not saying your workflow is chaotic... I'm just bringing a linter and a helmet.
openclaw tui - ws://127.0.0.1:18789 - agent main - session tui-1f61f07c-0e08-485f-b16c-6ff24c8e6de5
session agent:main:tui-1f61f07c-0e08-485f-b16c-6ff24c8e6de5
new session: tui-1f61f07c-0e08-485f-b16c-6ff24c8e6de5
```

Then, tell OpenClaw what your API key is:

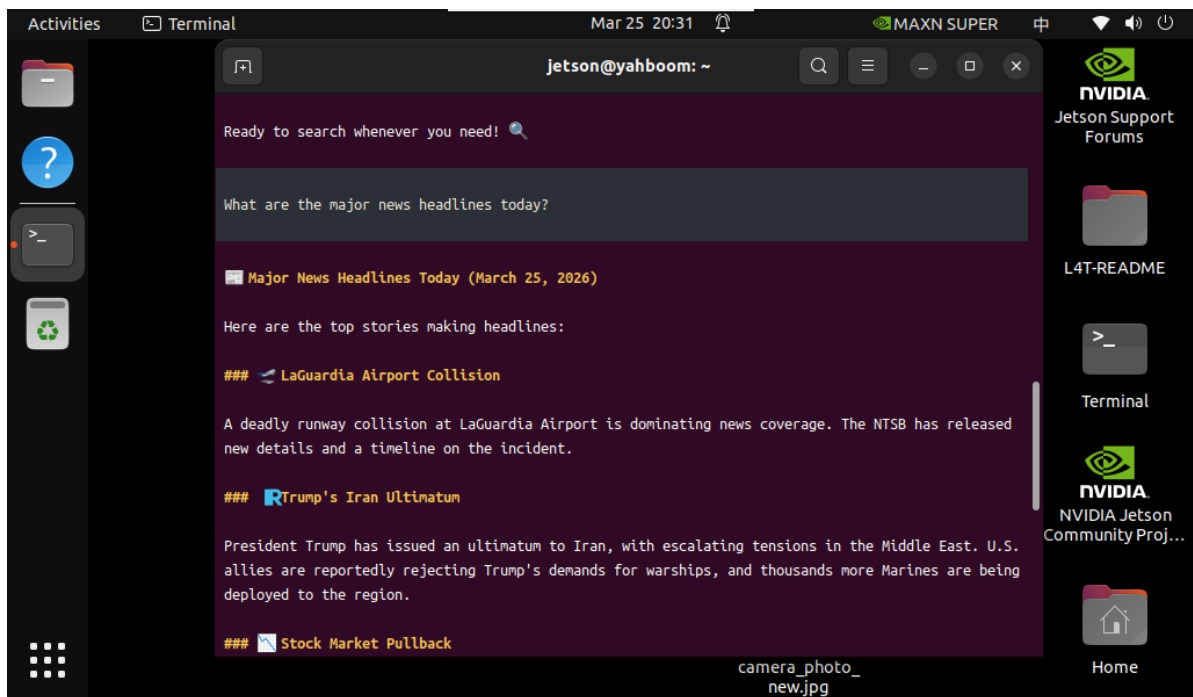
"This is my tavily key: tvlxxxxxxxxxxxxxxxxxxxxx"



4.2 Test Basic Search

In the TUI, enter a query that requires up-to-date information—for example:

what are the major news headlines today?



OpenClaw will:

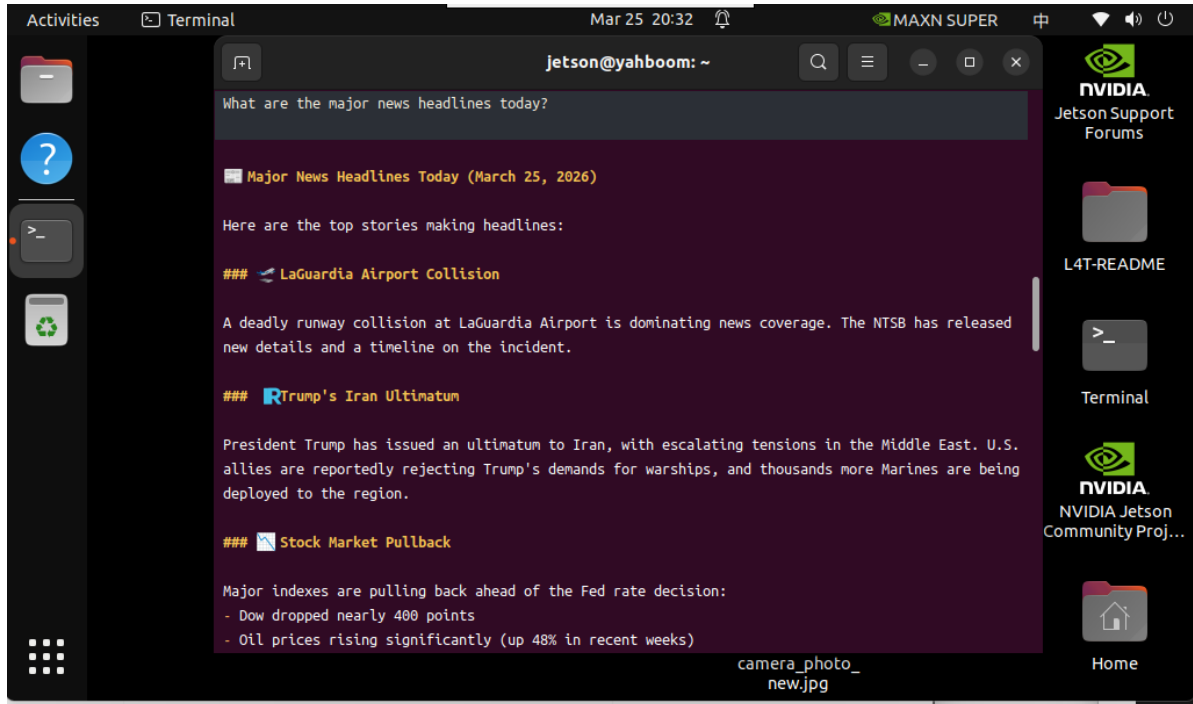
1. Identify when the latest information needs to be searched for.
2. Call the Tavily API to perform the search.
3. Retrieve the search results.
4. Answer your question based on the search results.

Advanced Usage Tips

Tip 1: Restricting Search Sources

If you only want to search for content on specific websites:

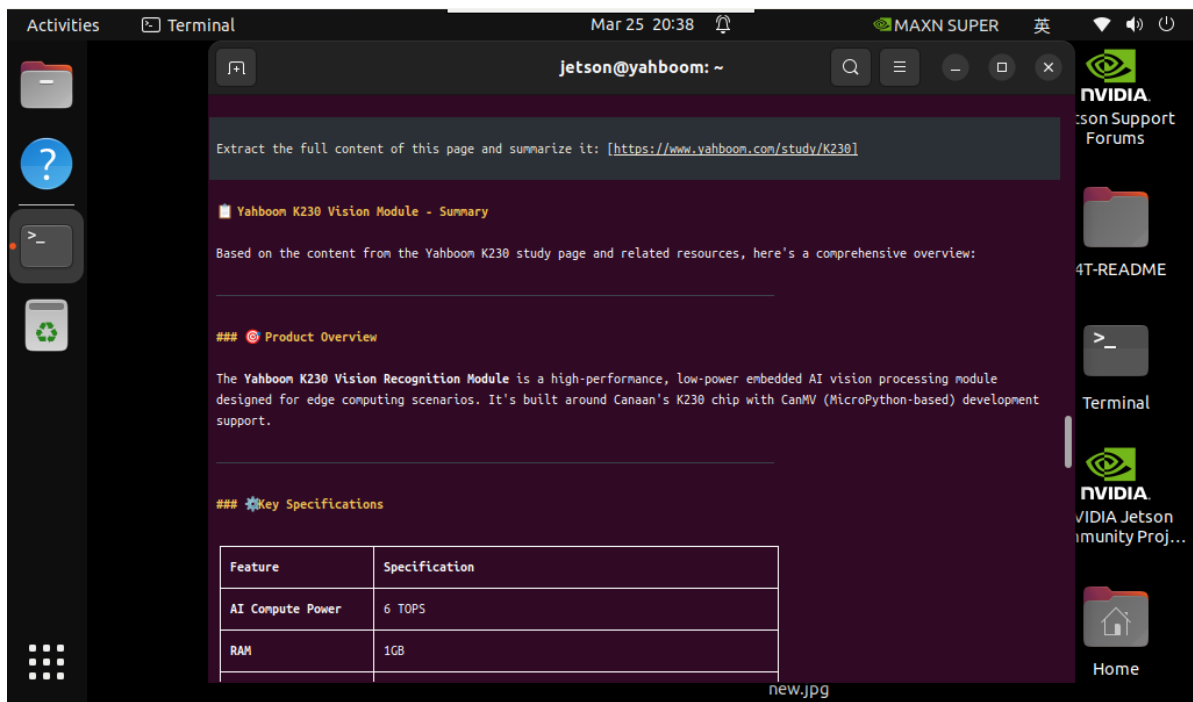
```
search only within MDN documentation: CSS Flexbox usage
```



Tip 2: Retrieving Original Content

If you need to retrieve the full content of a webpage:

```
Extract the full content of this page and summarize it:  
[https://www.yahboom.com/study/k230]
```



Frequently Asked Questions

Question 1: Search returns "Invalid API Key" error

Error Message:

```
Error: Invalid API key
```

Possible Causes:

1. The API key was entered incorrectly.
2. The environment variable was not set correctly.
3. The API key has expired.

Solution:

1. Check if the API key was copied correctly:

```
echo $TAVILY_API_KEY
```

2. Reset the environment variable:

```
export TAVILY_API_KEY=tvly-your-api-key-here
```

3. If the key has expired, log in to the official Tavily website to obtain a new one.

Question 2: Search yields no results

Symptoms:

After sending a search request, the AI responds with "Unable to find relevant information."

Possible Causes:

1. Network connection issues.

2. The Tavily service is temporarily unavailable.
3. Search keywords are too vague

Solution:

1. Check network connection:

```
ping tavily.com
```

2. Try more specific search keywords
3. Try again later

Issue 3: Search is very slow

Symptoms:

Search requests take a long time to return results

Possible Causes:

1. Deep search mode is being used
2. High network latency
3. High load on Tavily servers

Solution:

1. Use fast search mode:

```
Fast Search: [Your Query]
```

2. Check network speed:

```
curl -o /dev/null -s -w "%{time_total}\n" https://tavily.com
```

3. If it remains slow, try switching to a different network environment

Issue 4: Quota exhausted

Error Message:

```
Error: API quota exceeded
```

Possible Cause:

The free quota (1,000 requests per month) has been exhausted

Solution:

1. Log in to the Tavily console to check usage
2. Wait for the quota to reset next month

Issue 5: Skill installation failed

Error Message:

```
Error: Failed to install skill
```

Possible Causes:

- 1. Network connection issues
- 2. Outdated npm version
- 3. Insufficient permissions (rate-limited because you are not logged in)

Solution:

- 1. Check network connection
- 2. Update npm:

```
npm install -g npm@latest
```

- 3. Install with sudo privileges (if necessary):

```
sudo npx clawhub@latest install tavily-web-search
```

Search Mode Comparison

Mode	Speed	Coverage	Use Case	Example
Basic Mode	1-2 seconds	Good	Quick fact-checking, simple searches	"What is the weather like today?"
Advanced Mode	5-10 seconds	Excellent	Complex research, comprehensive answers	"Compare the pros and cons of two technologies"
News Mode	2-3 seconds	Last 7 days	Current events, Breaking news	"What's the latest tech news?"

Best Practices

1. Use Search Quotas Wisely

- ☒ Prioritize using the "Quick Search" mode.
- ☒ Avoid repeatedly searching for the same content.
- ☒ Leverage the AI's memory capabilities to avoid redundant queries.
- ☒ Do not frequently test the search functionality.

2. Optimize Search Keywords

- ☒ Use specific and precise questions.
- ☒ Include key information (time, location, subject).
- ☒ Describe your needs using natural language.
- ☒ Avoid overly vague queries.

3. Verify Search Results

- ☒ Cross-reference important information.
- ☒ Check the reliability of the search result sources.
- ☒ Pay attention to the timeliness of the information.
- ☒ Do not rely entirely on a single source.

4. Combine with Local Knowledge

- ☒ For static knowledge, prioritize using local models.
- ☒ For dynamic information, use web search.
- ☒ Combine both approaches to obtain the best answers.
- ☒ Do not use search for every single question.

Advanced Configuration

Configuration 1: Adjust Search Depth

In the OpenClaw configuration file, you can adjust the default search depth:

```
{
  "skills": {
    "entries": {
      "tavily-web-search": {
        "enabled": true,
        "apiKey": "tvly-your-api-key-here",
        "config": {
          "searchDepth": "basic",
          "maxResults": 10,
          "includeAnswer": true,
          "includeRawContent": false
        }
      }
    }
  }
}
```

Configuration Details:

- `searchDepth`: `basic` or `advanced`
- `maxResults`: The maximum number of results to return (1-10)
- `includeAnswer`: Whether to include an AI-generated answer summary
- `includeRawContent`: Whether to include the raw page content

Configuration 2: Set Domain Filters

If you only want to search for content on specific websites:

```
{
  "skills": {
    "entries": {
      "tavily-web-search": {
        "enabled": true,
        "apiKey": "tvly-your-api-key-here",
        "config": {
          "includeDomains": ["github.com", "stackoverflow.com"],
          "excludeDomains": ["ads.example.com"]
        }
      }
    }
  }
}
```

Configuration 3: Configuring News Mode

Set the default search mode to News Mode:

```
{
  "skills": {
    "entries": {
      "tavily-web-search": {
        "enabled": true,
        "apiKey": "tvly-your-api-key-here",
        "config": {
          "searchDepth": "basic",
          "topic": "news",
          "days": 7
        }
      }
    }
  }
}
```

Configuration Details:

- `topic`: `general` or `news`
- `days`: The time range (in days) for news searches (1-30)

Practical Use Case Examples

Scenario 1: Learning New Technologies

Search for the latest React tutorials suitable for beginners.

OpenClaw:

- Searches for the latest introductory React tutorials.
- Filters content from multiple sources to identify resources suitable for beginners.
- Provides links to tutorials along with brief descriptions.
- Offers learning recommendations.

Scenario 2: Shopping Decisions

Search for a detailed comparison between the iPhone 15 and iPhone 15 Pro, including price and performance.

OpenClaw:

- Searches for the latest product comparison information.
- Summarizes differences in price, performance, and features.
- Provides purchasing recommendations.
- Offers official purchase links.

Scenario 3: Travel Planning

Search for a comparison of transportation options from Beijing to Shanghai, including travel time, cost, and comfort levels.

OpenClaw:

- Searches for the latest transportation information.
- Compares options such as flights, high-speed trains, and buses.
- Provides price and travel time estimates.
- Offers recommended travel plans.

Scenario 4: Health Consultation

Search for natural remedies to alleviate insomnia, based on authoritative medical websites.

OpenClaw:

- Searches for information on authoritative medical websites
- Provides scientifically validated methods
- Outlines important precautions
- Recommends consulting a medical professional

Scenario 5: Programming Questions

You:

Search for how to implement asynchronous programming in Python, including code examples.

OpenClaw:

- Searches for the latest programming tutorials
- Provides detailed code examples
- Explains key concepts
- Offers best practice recommendations

Security & Privacy

Data Privacy

Tavily's Privacy Policy:

- ☒ Zero Data Retention: Search data is not permanently stored
- ☒ SOC 2 Certified: Complies with international security standards
- ☒ Prompt Injection Protection: Prevents malicious queries
- ☒ No Collection of Personally Identifiable Information (PII)

Security Recommendations

1. Protect API Keys

- Do not include keys in public code
- Rotate keys regularly
- Use environment variables or a secrets manager

2. Verify Search Results

- Check the credibility of information sources
- Cross-reference critical information
- Pay attention to the timeliness of the information

3. Responsible Use

- Adhere to Tavily's Terms of Use
- Do not misuse the search functionality
- Respect website content policies

Conclusion

Congratulations! You have successfully configured web search capabilities; OpenClaw can now access real-time information from the internet!

Next steps:

- 📖 Continue learning about other skill configurations
- 🎮 Explore more practical use cases
- 🔑 Dive deeper into OpenClaw's advanced features
- 💡 Unleash your creativity to build more applications

Tip: If you encounter any issues, you can run `openc1aw doctor` at any time to perform a diagnostic check, or visit the [Official Tavily Documentation](#) for further assistance.